

Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Internet of Things [IT-4050]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design to set up an IoT architecture with sensors, gateways, and a cloud platform for healthcare monitoring systems. (5)
 - A)
 - B) Develop a new business model based on the IoT value chain for connected healthcare services. (3)
 - C) Choose an IoT device that can monitor environmental data and explain. (2)
- 2) Write a code to develop a home automation system where a button press controls light and a fan using Raspberry Pi. (5)
 - A)
 - B) In a smart home automation environment, various devices (such as smart lights, thermostats, and security cameras) need to communicate effectively. These devices generate data and events that must be processed and responded to in real-time or near-real-time. A messaging system is crucial to enable this seamless communication. Analyze the suitable messaging protocol for the given scenario. (3)
 - C) A city employs IoT sensors in waste bins to monitor fill levels and optimize collection routes based on demand. Identify the key enhancements of the Web of Things for a given IoT application. (2)
- 3) In the context of industrial IoT (IIoT), Big Data analytics plays a crucial role in enhancing decision-making processes. This case study examines a manufacturing company that implements predictive maintenance strategies using Big Data analytics to improve operational efficiency and reduce downtime. Analyze the impact of Big Data analytics on decision-making in industrial IoT applications. (5)
 - A)
 - B) Consider a smart agriculture system that uses IoT devices to monitor environmental conditions, crop health, and resource usage across a large agricultural area. By utilizing APIs and data aggregation tools, the farm management team aims to combine data from various IoT devices into a unified dataset for analysis. List the tools to wring value from IoT data and explain. (3)
 - C) Consider the implementation of a Smart Health Monitoring System in a healthcare (2)

facility to manage chronic diseases such as diabetes and hypertension. Identify the objectives of the ubiquitous IoT application.

- 4) Consider a Smart Healthcare Monitoring System (SHMS) designed to provide continuous monitoring of patients' vital signs and health conditions using IoT devices. The system aims to enhance patient care by providing real-time data to healthcare providers. Examine the impact of data latency on decision-making in IoT applications. (5)
- A)
- B) Identify a prototype for an IoT application that utilizes machine learning to predict agricultural yield based on sensor data. (3)
- C) The rise of the Internet of Things (IoT) has led to the proliferation of smart home devices, allowing homeowners to monitor and control their environment remotely. Infer on the data representation and visualization. (2)
- 5) Compare the effectiveness of traditional healthcare delivery methods versus home healthcare solutions in managing chronic diseases. What are the strengths and weaknesses of each approach? (5)
- A)
- B) List the safety standards that influence the design of consumer electronics with an example. (3)
- C) A smart home system is designed to enhance the comfort, convenience, and security of a residential environment. Given a scenario of a smart home system, identify the sensors and actuators that are necessary for monitoring and controlling the environment. (2)

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